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Java SE 27 & JDK 27

Class AsynchronousChannelProvider

java.lang.Object
    java.nio.channels.spi.AsynchronousChannelProvider

```
public abstract class AsynchronousChannelProvider
extends Object
```

Service-provider class for asynchronous channels.

An asynchronous channel provider is a concrete subclass of this class that has a zero-argument constructor and implements the abstract methods specified below. A given invocation of the Java virtual machine maintains a single system-wide default provider instance, which is returned by the `provider` method. The first invocation of that method will locate the default provider as specified below.

All
of the methods in this class are safe for use by multiple concurrent threads.

Since:

1.7

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	AsynchronousChannelProvi	Initializes a new instance of this class.

Method Summary

All Methods | **Static Methods**

Modifier and Type	Method	Description
abstract AsynchronousChannelG	openAsynchronousChannelG (int nThreads, ThreadFactory threadFact	Constructs a new asynchronous channel group with a fixed thread pool.
abstract AsynchronousChannelG	openAsynchronousChannelG (ExecutorService executo int initialSize)	Constructs a new asynchronous channel group with the given thread pool.

abstract AsynchronousServerSo	openAsynchronousServerSo(AsynchronousChannelGrou	Opens an asynchronous server-socket channel.
abstract AsynchronousSocketCh	openAsynchronousSocketCh(AsynchronousChannelGrou	Opens an asynchronous socket channel.
static AsynchronousChannelP	provider()	Returns the system-wide default asynchronous channel provider for this invocation of the Java virtual machine.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Details

AsynchronousChannelProvider

```
protected AsynchronousChannelProvider()
```

Initializes a new instance of this class.

Method Details

provider

```
public static AsynchronousChannelProvider provider()
```

Returns the system-wide default asynchronous channel provider for this invocation of the Java virtual machine.

The first invocation of this method locates the default provider object as follows:

1. If the system property `java.nio.channels.spi.AsynchronousChannelProvider` is defined then it is taken to be the fully-qualified name of a concrete provider class. The class is loaded and instantiated; if this process fails then an unspecified error is thrown.

2. If a provider class has been installed in a jar file that is visible to the system class loader, and that jar file contains a provider-configuration file named `java.nio.channels.spi.AsynchronousChannelProvider` in

the resource directory `META-INF/services`, then the first class name specified in that file is taken. The class is loaded and instantiated; if this process fails then an unspecified error is thrown.

3. Finally, if no provider has been specified by any of the above means then the system-default provider class is instantiated and the result is returned.

Subsequent invocations of this method return the provider that was returned by the first invocation.

Returns:

The system-wide default AsynchronousChannel provider

openAsynchronousChannelGroup

```
public abstract AsynchronousChannelGroup openAsynchronousChannelGroup
(int nThreads,
 ThreadFactory threadFactory)
                                                                    throws
IOException
```

Constructs a new asynchronous channel group with a fixed thread pool.

Parameters:

`nThreads` - The number of threads in the pool

`threadFactory` - The factory to use when creating new threads

Returns:

A new asynchronous channel group

Throws:

`IllegalArgumentException` - If `nThreads <= 0`

`IOException` - If an I/O error occurs

See Also:

`AsynchronousChannelGroup.withFixedThreadPool(int, ThreadFactory)`

openAsynchronousChannelGroup

```
public abstract AsynchronousChannelGroup openAsynchronousChannelGroup
(ExecutorService executor,
 int initialSize)
                                                                    throws
IOException
```

Constructs a new asynchronous channel group with the given thread pool.

Parameters:

`executor` - The thread pool

`initialSize` - A value `>=0` or a negative value for implementation specific default

Returns:

A new asynchronous channel group

Throws:

`IOException` - If an I/O error occurs

See Also:

`AsynchronousChannelGroup.withCachedThreadPool(ExecutorService, int)`

openAsynchronousServerSocketChannel

```
public
abstract AsynchronousServerSocketChannel openAsynchronousServerSocketChannel
(AsynchronousChannelGroup group)

throws IOException
```

Opens an asynchronous server-socket channel.

Parameters:

`group` - The group to which the channel is bound, or `null` to bind to the default group

Returns:

The new channel

Throws:

`IllegalChannelGroupException` - If the provider that created the group differs from this provider

`ShutdownChannelGroupException` - The group is shutdown

`IOException` - If an I/O error occurs

openAsynchronousSocketChannel

```
public
abstract AsynchronousSocketChannel openAsynchronousSocketChannel
(AsynchronousChannelGroup group)

throws IOException
```

Opens an asynchronous socket channel.

Parameters:

`group` - The group to which the channel is bound, or `null` to bind to the default group

Returns:

The new channel

Throws:

`IllegalChannelGroupException` - If the provider that created the group differs from this provider

`ShutdownChannelGroupException` - The group is shutdown

`IOException` - If an I/O error occurs

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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